

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Mid
Semester: Fall 2023

Duration: 1 Hour 10 min
Full Marks: 25

CSE 470: Software Engineering
SET B

Name:

ID:

Section:

Q1. [CO1] Consider a situation where Google has taken a project to develop an AI-based chat Application. The user should be able to send messages to other users. They should also be able to send images and file attachments along with text messages. The chat application supports multiple languages and integrates chatbots to automate common tasks and provide instant responses. Furthermore, it must maintain high availability to ensure it is accessible to users 24/7, while also giving priority to ensuring data security and user privacy. Now, Sundar Pichai has chosen you as the team lead for the project. He allows you to choose your teammates as per your preference. Now, you are familiar with the project requirements and it is a medium-scale project. However, Sundar Pichai has targeted to have the end product within 10 versions. Understanding the importance of the project, as a team leader you have divided the project into 60 modules and assigned 11 software engineers to work on that project.

- A. **What** software development process model will you follow in the given scenario? Justify your answer. [3 marks]
- B. As a team lead, **what** would be the process to check the project progress according to your framework methodology? [3 marks]
- C. Sundar Pichai later informs you that because of the competitive tech-market they want the product ready as soon as possible. He divides the whole timeline into four divisions and asks you to give an update after each period, **Which** development approach will you follow now? [3 marks]
- D. **What** are the functional and non-functional requirements in the above scenario? [4 marks]

Q2. [CO2] You are tasked with designing a class diagram for an Autonomous Agriculture Management System, which is used to oversee and automate various aspects of a large-scale agricultural operation. Here are the key components and functionalities of the system:

- **Farm:** The system manages multiple farms, each with unique attributes such as location, size, and crop type.
- **Crop Management:** The system tracks information about the crops grown on each farm, including crop type, planting date, harvest date, and growth status.
- **Weather Data:** The system collects and processes real-time weather data, including temperature, humidity, precipitation, and wind speed, to make informed decisions about crop management.
- **Autonomous Vehicles:** The system includes autonomous vehicles (e.g., drones, tractors) equipped with various sensors and tools for tasks such as planting, harvesting, and pest control.

- **Irrigation System:** The irrigation system is responsible for managing water resources, including sensors to measure soil moisture and automated watering mechanisms.
- **Pest Control:** The system monitors for pests and diseases in crops and uses data to apply targeted pest control measures.
- **Crop Health Analysis:** Analyze data from sensors to assess crop health, detect anomalies, and recommend appropriate actions.
- **Inventory Management:** Keep track of agricultural equipment, spare parts, and other resources required for farm operations.
- **Harvest and Yield Analysis:** Record data on harvested crops and calculate yields for each crop type.
- **Resource Allocation:** Allocate resources (e.g., vehicles, labor, water) efficiently based on real-time data, weather forecasts, and crop conditions.
- **User Roles:** The system supports different user roles, such as farmers, agricultural engineers, and data analysts, each with specific access and functionality.
- **Reporting and Data Visualization:** Provide tools for generating reports and visualizing data, including charts and graphs for farm performance.

As you design the class diagram for this scenario, consider the relationships between these classes, the encapsulation of data and methods, and the system's overall architecture.

[8 marks]

Q3. [CO3] You are tasked with designing a data processing system for a large e-commerce platform. The system needs to handle a high volume of incoming product data from various suppliers, perform data transformation and enrichment, and then route the processed data to different downstream services, including inventory management, pricing, and product catalog updates. To achieve this, you will definitely require various filtering and be able to process transaction-based data.

A. **What** kind of software pattern you will use to solve the above scenario? Explain one advantage and one disadvantage of your software pattern in this scenario. [1+2 marks]

B. **What** is one disadvantage of using a Monolithic software architecture in this scenario? [1 marks]

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